

GAINESVILLE 2 NE, AL, USA

WMO: 750030

Lat: 32.8369N

Lon: 88.1374W

Elev: 33

StdP: 100.93

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 63892

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -5.3	(c) -3.0	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 6.5	(k) 3.4	(l) 5.8	(m) 4.0	(n) 2.5	(o) 0	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.9	(c) 35.2	(d) N/A	(e) 34.0	(f) N/A	(g) 32.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.0	(p) 340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.4	(b) 4.7	(c) 4.0	(d) DB	(e) -8.2	(f) 36.4	(g) 2.4	(h) 1.3	(i) -9.9	(j) 37.4	(k) -11.2	(l) 38.2	(m) -12.6	(n) 38.9	(o) -14.3	(p) 39.8	(q) N/A
(a) 5.4	(b) 4.7	(c) 4.0	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	7.0	9.6	13.8	17.6	21.9	26.2	26.9	27.0	24.5	18.3	11.9	9.5
	DBStd	8.19	5.80	5.36	5.09	3.81	3.20	1.99	1.72	2.01	2.88	4.48	4.61	5.36
	HDD10.0	333	128	67	24	2	0	0	0	0	0	1	32	78
	HDD18.3	1356	351	248	154	58	11	0	0	0	1	57	198	278
	CDD10.0	3217	37	56	141	229	370	486	524	527	436	259	89	63
	CDD18.3	1199	1	3	13	36	122	236	266	268	187	57	6	4
	CDH23.3	10773	2	16	100	318	1023	2279	2459	2469	1576	490	31	9
	CDH26.7	4493	0	1	12	48	341	1058	1084	1131	676	139	3	0
(14) Wind	WSAvg	1.8	2.2	2.1	2.2	2.0	1.5	1.6	1.4	1.4	1.4	1.7	1.7	2.0
(15) Precipitation	PrecAvg	1319	143	136	144	119	89	95	117	92	94	77	102	130
	PrecMax	1841	342	354	338	468	250	220	297	257	229	208	369	490
	PrecMin	908	29	39	37	2	3	16	29	9	17	0	17	48
	PrecStd	216	62	70	71	86	54	49	69	57	55	48	63	73
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.4	25.4	28.2	29.4	32.7	36.1	35.6	37.4	35.3	31.7	26.4	24.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	21.0	22.9	26.1	28.1	31.3	34.7	34.1	35.3	33.7	30.0	23.9	22.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	18.7	20.8	24.0	26.9	30.1	33.4	33.1	33.8	32.4	28.2	22.1	20.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	16.4	18.7	22.2	25.3	28.8	32.0	32.0	32.3	30.9	26.2	20.1	18.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	11.4	11.7	12.3	11.5	11.0	9.9	10.2	11.2	12.4	12.7	10.3
		MCDBR	12.5	13.3	13.5	13.6	13.0	12.8	11.2	12.4	12.9	14.0	13.9	11.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.326	0.333	0.364	0.400	0.437	0.482	0.526	0.524	0.439	0.373	0.346	0.324
	taud		2.515	2.496	2.423	2.314	2.256	2.161	2.062	2.063	2.286	2.454	2.473	2.537
	Ebn at Noon		893	923	914	890	856	814	775	771	827	865	860	876
	Edh at Noon		85	96	111	129	138	151	166	164	126	99	88	79
(44) All-Sky Solar Radiation	RadAvg		2.70	3.23	4.37	5.51	6.08	6.30	6.04	5.58	4.96	4.12	3.05	2.31
	RadStd		0.22	0.44	0.40	0.43	0.42	0.48	0.36	0.37	0.40	0.52	0.31	0.23

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+143	+96

Nomenclature: See separate page